

COURSE CODE/TITLE: 010-CPE21S1 - Data Structures and Algorithms

SECTION: CPE21S1

DATE PERFORMED: July 28 2026

DATE SUBMITTED: July 28 2026

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ACTIVITY NO. 4.1 Stacks

1. PROCEDURE OUTPUT

What procedures did you perform?

Table 4-1. Output of ILO A

```
1 //Tests the push, empty, size, pop, and top methods of the stack library.
2 #include <iostream>
3 #include <stack> // Calling Stack from the STL
4 using namespace std;
5 int main() {
6
7     stack<int> newStack;
8     newStack.push(3); //Adds 3 to the stack
9     newStack.push(8);
10    newStack.push(15);
11
12    // returns a boolean response depending on if the stack is empty or not
13    cout << "Stack Empty? " << newStack.empty() << endl;
14    // returns the size of the stack itself
15    cout << "Stack Size: " << newStack.size() << endl;
16    // returns the topmost element of the stack
17    cout << "Top Element of the Stack: " << newStack.top() << endl;
18    // removes the topmost element of the stack
19    newStack.pop();
20    cout << "Top Element of the Stack: " << newStack.top() << endl;
21    cout << "Stack Size: " << newStack.size() << endl;
22    return 0;
23 }
```

```
C:\Users\royga\CLionProjects\untitled1\keyufg.exe
is stack empty? 1
The top of the element is 15
is stack empty? 0
The size of the stack is 3
Removing elements from the stack
The top of the element is 8
The top of the element is 3
```

Code Analysis:

On this part of the activity, we have been able to demonstrate the use of the stack and its functions such as the push, pop, top, size, isEmpty, isFull, and all of which are important part of stack

Table 4-2. Output of ILO B.1

Code:

```

1  ● #include <iostream>
2
3  const size_t maxCap = 100;
4  int stack[maxCap];
5  int top = -1;
6  int newData, choice, i;
7
8  ↵ void push();
9  ↵ void pop();
10 ↵ void Top();
11 ↵ bool isEmpty();
12 bool isFull();
13 ↵ void prinAll();
14
15 ▶ ∨ int main() {
16     //=====
17     std::cout << "Enter the maximum number of elements (1-100): ";
18     std::cin >> i;
19     //=====
20     // Validate the user input
21
22     if (i <= 0 || i > maxCap) {
23         std::cout << "Invalid! Maximum allowed is " << maxCap << ".\n";
24         return 0;
25     }
26     //=====
27     while (true) {
28         std::cout << "\n==== STACK OPERATIONS =====\n";
29         std::cout << "1. PUSH\n" << "2. POP\n" << "3. TOP\n" << "4. isEmpty\n" << "5. PRINT CONTENTS\n" << "6. EXIT\n" << "Choose your choice: ";
30         //=====
31         std::cin >> choice;
32         switch (choice) {
33             case 1:
34                 push();
35                 break;
36             case 2:
37                 pop();
38                 break;
39             //=====
40             case 3:

```

```

41         Top();
42         break;
43         //=====
44     case 4:
45         if (isEmpty())
46             std::cout << "Stack is Empty.\n";
47         else
48             std::cout << "Stack is NOT Empty.\n";
49         break;
50         //=====
51     case 5:
52         std::cout<<"Print out all the contents of the stack.\n";
53         prinAll();
54         break;
55         //=====
56     case 6:
57         std::cout << "Exiting program.\n";
58         return 0;
59         break;
60         //

```

```

61         return 0;
62         break;
63         //=====
64     default:
65         std::cout << "Invalid choice!\n";
66         break;
67     }
68     return 0;
69 }
70 //=====
71 //=====
72 //checks if the stack is empty
73 → bool isEmpty() {
74     if (top== -1) return true;
75     return false;
76 }
77 //=====
78 //adds new content to the very top

```

```

78 → void push() {
79     if (top==i-1) {
80         std::cout << "Stack Overflow! Cannot push more elements.\n";
81         return;
82     }
83     std::cout << "Enter value to push: ";
84     std::cin >> newData;
85     stack[++top] = newData;
86     std::cout << newData << " pushed into the stack.\n";
87 }
88 //=====
89 //removes the topmost content
90 → void pop() {
91     if (isEmpty()) {
92         std::cout << "Stack Underflow! Stack is empty.\n";
93         return;
94     }
95     std::cout << "Popping: " << stack[top] << std::endl;
96     top--;
97 }
98 //=====

```

```

//shows the latest and last in
100 → void Top() {
101     if (isEmpty()) {
102         std::cout << "Stack Underflow! Stack is empty.\n";
103         return;
104     }
105     std::cout << "Top element: " << stack[top] << std::endl;
106 }
107 //=====
108 // to print all the contents of the stack
109 //checking if the stack is empty
110 → void prinAll() {
111     if (isEmpty()) {
112         std::cout << "Stack is empty.\n";
113         return;
114     }
115     //literal printing of the stack
116     std::cout << "Stack elements (top to bottom):\n";
117     for (int j = top; j >= 0; --j) {
118         std::cout << "|" << stack[j] << "|" << std::endl;
119     }

```

Output:

```
===== STACK OPERATIONS =====
```

```
1. PUSH
```

```
2. POP
```

```
3. TOP
```

```
4. isEmpty
```

```
5. PRINT CONTENTS
```

```
6. EXIT
```

```
Choose your choice:1
```

```
Enter value to push:5
```

```
5 pushed into the stack.
```

```
===== STACK OPERATIONS =====
```

1. PUSH
2. POP
3. TOP
4. isEmpty
5. PRINT CONTENTS
6. EXIT

Choose your choice:1

Enter value to push:10

10 pushed into the stack.

```
===== STACK OPERATIONS =====
```

1. PUSH
2. POP
3. TOP
4. isEmpty
5. PRINT CONTENTS
6. EXIT

Choose your choice:1

Enter value to push:15

15 pushed into the stack.

```
===== STACK OPERATIONS =====
```

1. PUSH
2. POP
3. TOP
4. isEmpty
5. PRINT CONTENTS
6. EXIT

Choose your choice:1

Enter value to push:20

20 pushed into the stack.

```
===== STACK OPERATIONS =====
```

1. PUSH
2. POP
3. TOP
4. isEmpty
5. PRINT CONTENTS
6. EXIT

Choose your choice:3

Top element: 20


```
===== STACK OPERATIONS =====
```

```
1. PUSH
```

```
2. POP
```

```
3. TOP
```

```
4. isEmpty
```

```
5. PRINT CONTENTS
```

```
6. EXIT
```

```
Choose your choice:4
```

```
Stack is NOT Empty.
```

```
===== STACK OPERATIONS =====
```

```
1. PUSH
```

```
2. POP
```

```
3. TOP
```

```
4. isEmpty
```

```
5. PRINT CONTENTS
```

```
6. EXIT
```

```
Choose your choice:5
```

```
Print out all the contents of the stack.
```

```
Stack elements (top to bottom):
```

```
|20|
```

```
|15|
```

```
|10|
```

```
|5|
```

```
===== STACK OPERATIONS =====
```

1. PUSH
2. POP
3. TOP
4. isEmpty
5. PRINT CONTENTS
6. EXIT

Choose your choice:2

Popping: 20

```
===== STACK OPERATIONS =====
```

1. PUSH
2. POP
3. TOP
4. isEmpty
5. PRINT CONTENTS
6. EXIT

Choose your choice:5

Print out all the contents of the stack
Stack elements (top to bottom):

|15|
|10|
|5|

Code Analysis:

In this part of the activity, we are tasked to use an array for a stack, where we also apply what we did on the first task, but here we must be able to remove and add and show the final content of the stack if it is not empty. The use of **Push()**, the function that adds a new content to the very top of the array, was helpful in the creation of one of the most important functions to have the contents of the stack. Both the **isEmpty**, and **isFull**, is important at a certain scenario to know if the array stack is already full or empty. The **Top()** is basically outputting or showing the most latest or the content that is inputted the last. Then we have the **Pop()** one of the important functions for it removes the content or removes an element of the array stack from the stack itself. Lastly, the student made function of **prinAll()**, this just basically shows the contents of the entire stack without disturbing the distribution of the contents.

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

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